

Q-1 : Differentiate $\log_a x$ by the definition.

Solution : let, $f(x) = \log_a x$

$$f(x+h) = \log_a(x+h) = \log_e x \cdot \log_a e$$

$$= \ln x \cdot \log_a e$$

$$f(x+h) = \ln(x+h) \cdot \log_a e$$

Now by the definition, we have,

$$\frac{d}{dx} \{f(x)\} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\therefore \frac{d}{dx} (\log_a x) = \lim_{h \rightarrow 0} \frac{\ln(x+h) \cdot \log_a e - \ln x \cdot \log_a e}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\log_a e \{ \ln(x+h) - \ln x \}}{h}$$

$$= \log_a e \lim_{h \rightarrow 0} \frac{\ln(1 + \frac{h}{x})}{h}$$

$$= \log_a e \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{h}{x} - \frac{h^2}{2x^2} + \frac{h^3}{3x^3} + \dots \infty \right]$$

$$= \log_a e \lim_{h \rightarrow 0} \frac{h}{h} \left[\frac{1}{x} - \frac{h}{2x^2} + \frac{h^2}{3x^3} + \dots \infty \right]$$

$$= \frac{1}{x} \log_a e$$

$$\text{Hence, } \frac{d}{dx} (\log_a x) = \frac{1}{x} \log_a e \text{ (Ans.)}$$

Q-16 : Differentiate $y = x^3 \left(\frac{x^2+4}{x^2+3} \right)^{1/2}$ with respect to x .

Solution: Given that,

$$y = x^3 \left(\frac{x^2+4}{x^2+3} \right)^{1/2}$$

Taking $\ln$ on both sides we have,

$$\ln y = \ln x^3 + \ln \left(\frac{x^2+4}{x^2+3} \right)^{1/2}$$

$$= 3 \ln x + \frac{1}{2} \ln(x^2+4) - \frac{1}{2} \ln(x^2+3)$$

Now diff. w. r. to x we have,

$$\frac{1}{y} \cdot \frac{dy}{dx} = 3 \cdot \frac{1}{x} + \frac{1}{2} \cdot \frac{1}{x^2+4} \cdot 2x - \frac{1}{2} \cdot \frac{1}{x^2+3} \cdot 2x$$

$$\therefore \frac{dy}{dx} = y \left[\frac{3}{x} + \frac{x}{x^2+4} - \frac{x}{x^2+3} \right]$$

$$\text{Hence, } \frac{d}{dx} \left\{ x^3 \left(\frac{x^2+4}{x^2+3} \right)^{1/2} \right\} = x^3 \left(\frac{x^2+4}{x^2+3} \right)^{1/2} \left[\frac{3}{x} + \frac{x}{x^2+4} - \frac{x}{x^2+3} \right]$$

Q-11 : Differentiate $\tan^{-1} \frac{2x}{1-x^2}$ with respect to $\sin^{-1} \frac{2x}{1+x^2}$

Solution : Let, $y = \tan^{-1} \frac{2x}{1-x^2}$

$$= \tan^{-1} \frac{2 \tan \theta}{1 - \tan^2 \theta}$$
$$= \tan^{-1} \tan 2\theta$$
$$= 2\theta$$
$$= 2 \tan^{-1} x$$

Let,
 $x = \tan \theta$
 $\therefore \theta = \tan^{-1} x$

Now diff. w. r. to x we get,

$$\frac{dy}{dx} = \frac{2}{1+x^2}$$

Again, let, $z = \sin^{-1} \frac{2x}{1+x^2}$

$$= \sin^{-1} \frac{2 \tan \theta}{1 + \tan^2 \theta}$$
$$= \sin^{-1} \sin 2\theta$$
$$= 2\theta$$
$$= 2 \tan^{-1} x$$

Let,
 $x = \tan \theta$
 $\therefore \theta = \tan^{-1} x$

Now diff. w. r. to x we get,

$$\frac{dz}{dx} = \frac{2}{1+x^2}$$

Q-21: If, $y = (\cos x)^{(\cos x)^{(\cos x) \dots}}$

Then find $\frac{dy}{dx}$.

Solution: Given that,

$$y = (\cos x)^{(\cos x)^{(\cos x) \dots}} \quad \dots \textcircled{1}$$

Taking $\ln$ on both sides we have,

$$\begin{aligned} \ln y &= (\cos x)^{(\cos x)^{(\cos x) \dots}} \ln (\cos x) \\ &= y \ln (\cos x) \quad [\text{by } \textcircled{1}] \end{aligned}$$

Now diff. w. r. to x we get,

$$\frac{1}{y} \cdot \frac{dy}{dx} = y \cdot \frac{1}{\cos x} \cdot (-\sin x) + \ln \cos x \cdot \frac{dy}{dx}$$

$$\text{on, } \left(\frac{1}{y} - \ln \cos x \right) \frac{dy}{dx} = -y \tan x$$

$$\text{on, } \frac{dy}{dx} = \frac{-y \tan x}{\frac{1}{y} - \ln \cos x}$$

$$\therefore \frac{dy}{dx} = \frac{-y^2 \tan x}{1 - y \ln \cos x} \quad (\text{Ans})$$

Q-25: If, $y = \sqrt{\sin x \sqrt{\sin x \sqrt{\sin x} \dots \infty}}$ then find $\frac{dy}{dx}$.

Solution : Given that,

$$y = \sqrt{\sin x \sqrt{\sin x \sqrt{\sin x} \dots \infty}} \quad \dots \textcircled{1}$$

Now squaring both sides we have,

$$y^2 = \sin x \sqrt{\sin x \sqrt{\sin x \sqrt{\sin x} \dots \infty}}$$

$$\text{or, } y^2 = y \sin x \quad [\text{by } \textcircled{1}]$$

$$\text{or, } y = \sin x$$

Now diff. w. r. to x we get,

$$\frac{dy}{dx} = \cos x \quad (\text{Ans})$$

(Ans)

Q-27: If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \infty}}}$ then find dy/dx .

Solution: Given that,

$$y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \infty}}} \quad \text{--- ①}$$

Now squaring both sides we have

$$y^2 = \sin x + \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \infty}}}$$

$$\text{or, } y^2 = \sin x + y \quad [\text{by ①}]$$

Now diff. w.r. to x we get,

$$2y \frac{dy}{dx} = \cos x + \frac{dy}{dx}$$

$$\text{or, } (2y - 1) \frac{dy}{dx} = \cos x$$

$$\text{or, } \frac{dy}{dx} = \frac{\cos x}{2y - 1} \quad (\text{Ans})$$

Q-30: If, $y = x^2 + \frac{1}{x^2} + \frac{1}{x^2} + \dots \infty$

then find $\frac{dy}{dx}$.

Solution: Given that, $y = x^2 + \frac{1}{x^2} + \frac{1}{x^2} + \dots \infty$

$$y = x^2 + \frac{1}{x^2} + \frac{1}{x^2} + \dots \infty$$

$$\text{on, } y = x^2 + \frac{1}{x^2} + \frac{1}{x^2} + \frac{1}{x^2} + \dots \infty \quad (1)$$

$$\text{on, } y = x^2 + \frac{1}{y} \quad [\text{by } (1)]$$

$$\text{on, } y = \frac{x^2 y + 1}{y}$$

$$\text{on, } y^2 = x^2 y + 1$$

Now diff. w.r. to x we get,

$$2y \cdot \frac{dy}{dx} = x^2 \frac{dy}{dx} + y \cdot 2x$$

$$\text{on, } (2y - x^2) \frac{dy}{dx} = 2xy$$

$$\text{on, } \frac{dy}{dx} = \frac{2xy}{2y - x^2} \quad (\text{Ans})$$

Successive Differentiation

পর্যায়ক্রমিক অন্তরীকরণ

• If $y = f(x)$ be any function then,

1st order derivatives, $\frac{dy}{dx} = Dy = y_1 = y' = f'(x)$

2nd order derivative, $\frac{d^2y}{dx^2} = D^2y = y_2 = y'' = f''(x)$

3rd order derivative, $\frac{d^3y}{dx^3} = D^3y = y_3 = y''' = f'''(x)$

Similarly,

nth order derivative, $\frac{d^ny}{dx^n} = D^ny = y_n = y^n = f^n(x)$

Q-34: Find the n^{th} derivative of the function

$$y = f(x) = x^n.$$

Solution: Given that,

$$y = f(x) = x^n$$

Now diff w.r.t. to x successively,

$$y_1 = n x^{n-1}$$

$$y_2 = n(n-1) x^{n-2}$$

$$y_3 = n(n-1)(n-2) x^{n-3}$$

$$\dots \dots \dots$$

Similarly the n^{th} derivative,

$$y_n = n(n-1)(n-2) \dots \dots \dots \{n-(n-1)\} x^{n-n}$$

$$= n(n-1)(n-2) \dots \dots \dots 3 \cdot 2 \cdot 1 \cdot x^0$$

$$= 1 \cdot 2 \cdot 3 \dots \dots (n-2)(n-1)n$$

$$= n \text{ or } n! \quad (\text{Ans})$$

Leibnitz's Theorem

If u and v are the function of x then the n^{th} derivative of the product of u and v is given by,

$$(uv)_n = u_n v + {}^n C_1 u_{n-1} v_1 + {}^n C_2 u_{n-2} v_2 + \dots + {}^n C_r u_{n-r} v_r + \dots + u v_n$$

where the suffices of u and v indicates the order of definition with respect to x , which is called Leibnitz theorem.

Q-39: If $x = \sin\left(\frac{1}{m} \ln y\right)$ then using Leibnitz theorem, show that, $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0$

Solution: Given that,

$$x = \sin\left(\frac{1}{m} \ln y\right)$$

$$\text{on, } \sin^{-1} x = \frac{1}{m} \ln y$$

$$\text{on, } \ln y = m \sin^{-1} x$$

$$\text{on, } e^{\ln y} = e^{m \sin^{-1} x}$$

$$\text{on, } y = e^{m \sin^{-1} x} \quad [\because e^{\ln x} = x] \quad \dots \textcircled{1}$$

Now diff. w.r. to x we get,

$$y_1 = e^{m \sin^{-1} x} \cdot m \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\text{on, } y_1 = \frac{my}{\sqrt{1-x^2}} \quad [\text{by } \textcircled{1}]$$

$$\text{on, } y_1^2 = \frac{m^2 y^2}{1-x^2}$$

$$\text{on, } (1-x^2)y_1^2 = m^2 y^2 \quad \dots \textcircled{2}$$

Again, diff. ② w.r. to x we get,

$$(1-x^2) 2y_1 \cdot y_2 + y_1^2 (-2x) - m^2 2y \cdot y_1 = 0$$

$$\text{or, } (1-x^2) y_2 - x y_1^2 - m^2 y = 0 \quad [\text{div. by } 2y_1]$$

Now applying Leibnitz theorem we have,

$$(1-x^2) y_{n+2} + n c_1 (-2x) y_{n+1} + n c_2 (-2) y_n - [x y_{n+1} + n c_1 y_n] - m^2 y_n = 0$$

$$\text{or, } (1-x^2) y_{n+2} - 2n x y_{n+1} + \frac{n(n-1)}{2!} \cdot 2 y_n - x y_{n+1} - n y_n - m^2 y_n = 0$$

$$\text{or, } (1-x^2) y_{n+2} - 2n x y_{n+1} - x y_{n+1} - n^2 y_n - n y_n - m^2 y_n = 0$$

$$\text{or, } (1-x^2) y_{n+2} - (2n+1) x y_{n+1} - (n^2 + m^2) y_n = 0$$

Function

Definition:

Function: A function is a rule that associates, with each value of an independent variable x in a certain set exactly one value of another variable y , which is dependent.

Symbolically, we write,

$$y = f(x) \text{ or } y = \phi(x)$$

For example, $y = x^2$ or $y = 3x + 5$

Domain: The domain is the set of all possible x -values (inputs).

Range: The range is the set of all possible y -values (outputs).

Q-47: Find the domain and range of the function, $y = f(x) = x + 3$

Solution: Given that,

$$y = f(x) = x + 3 \dots \textcircled{1}$$

which is defined for all real values of x .

$$\therefore \text{Domain, } D_f = \{x : x \in \mathbb{R}\}$$

$$= (-\infty, \infty)$$

$$= \mathbb{R} \quad (\text{Ans})$$

Again for range, From $\textcircled{1}$ we have,

$$x = y - 3 \dots \textcircled{2}$$

which is defined for all real values of y .

$$\therefore \text{Range, } R_f = \{y : y \in \mathbb{R}\}$$

$$= (-\infty, \infty)$$

$$= \mathbb{R} \quad (\text{Ans})$$

Q-50: Find the domain and range of the function

$$y = f(x) = \sqrt{x-3}$$

Solution: Given that,

$$y = f(x) = \sqrt{x-3} \dots \textcircled{1}$$

which is defined when $x-3 \geq 0$

$$\Rightarrow x \geq 3$$

$$\begin{aligned} \therefore \text{Domain, } D_f &= \{x : x \geq 3\} \\ &= [3, \infty) \text{ (Ans)} \end{aligned}$$

Again for range,

when, $x = 3$ then $y = 0$

also, when, $x = \infty$ then $y = \infty$

$$\begin{aligned} \therefore \text{Range, } R_f &= \{y : y \geq 0\} \\ &= [0, \infty) \text{ (Ans)} \end{aligned}$$

Q. Find the domain and range of the function

$$y = f(x) = \frac{x-3}{2x+1}$$

Solution: Given that,

$$y = f(x) = \frac{x-3}{2x+1} \quad \text{--- (1)}$$

which is defined when $2x+1 \neq 0$
or, $x \neq -\frac{1}{2}$

$$\therefore \text{Domain, } D_f = \mathbb{R} - \left\{-\frac{1}{2}\right\} \quad (\text{Ans})$$

Again for range from (1) we have,

$$2xy + y = x - 3$$

$$\text{or, } 2xy - x = -y - 3$$

$$\text{or, } x(2y-1) = -(y+3)$$

$$\text{or, } x = \frac{-(y+3)}{(2y-1)}$$

$$\text{or, } x = \frac{y+3}{1-2y}$$

which is defined when $1-2y \neq 0$

$$\text{or, } y \neq \frac{1}{2}$$

$$\therefore \text{Range, } R_f = \mathbb{R} - \left\{ \frac{1}{2} \right\} \text{ (ans)}$$

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Q. Find the domain and range of the

$$\text{function, } y = f(x) = |x+1| + |x-1|$$

Solution: Given that,

$$\begin{array}{l} x+1=0 \Rightarrow x=-1 \\ x-1=0 \Rightarrow x=1 \end{array}$$

$$y = f(x) = |x+1| + |x-1|, \quad \begin{array}{c} \leftarrow x < -1 \quad -1 \leq x \leq 1 \quad x > 1 \\ \quad \quad \quad x = -1 \quad x = 1 \end{array}$$

$$= \begin{cases} -(x+1) - (x-1), & \text{when } x < -1 \\ (x+1) - (x-1), & \text{when } -1 \leq x \leq 1 \\ (x+1) + (x-1), & \text{when } x > 1 \end{cases}$$

$$= \begin{cases} -2x, & x < -1 \\ 2, & -1 \leq x \leq 1 \\ 2x, & x > 1 \end{cases}$$

$$\begin{aligned}
 \therefore \text{Domain, } D_f &= \{x: x < -1\} \cup \{x: -1 \leq x \leq 1\} \cup \{x: x > 1\} \\
 &= (-\infty, -1) \cup [-1, 1] \cup (1, \infty) \\
 &= (-\infty, \infty) \\
 &= \mathbb{R} \quad (\text{Ans})
 \end{aligned}$$

Again for range,

$$\text{when } x < -1 \text{ then } y = (2, \infty)$$

$$\text{when } -1 \leq x \leq 1 \text{ then } y = \{2\}$$

$$\text{when } x > 1 \text{ then } y = (2, \infty)$$

$$\begin{aligned}
 \therefore \text{Range, } R_f &= (2, \infty) \cup \{2\} \cup (2, \infty) \\
 &= [2, \infty) \quad (\text{Ans})
 \end{aligned}$$

Limit / সীমা

- ① Define limit of a function: $\lim_{x \rightarrow a} f(x) = L$.
- ② L.H.L = $\lim_{x \rightarrow a^-} f(x)$
- ③ R.H.L = $\lim_{x \rightarrow a^+} f(x)$
- ④ If L.H.L = R.H.L then the limit will be exist.
- ⑤ If L.H.L = R.H.L = $f(a)$ then the function will be continuous at $x = a$.
- ⑥ Define continuity of a function.

Q. Does the limit exist if $f(x) = \begin{cases} 3+2x, & -3/2 \leq x < 0 \\ 3-2x, & 0 \leq x < 3/2 \\ -3-2x, & x \geq 3/2 \end{cases}$

Does the limit exist at $x=0$?

Also test continuity of the function at $x=0$.

Solution: Given that,

$$f(x) = \begin{cases} 3+2x, & -3/2 \leq x < 0 \\ 3-2x, & 0 \leq x < 3/2 \\ -3-2x, & x \geq 3/2 \end{cases}$$

Now, L.H.L = $\lim_{x \rightarrow 0^-} f(x)$

$$= \lim_{x \rightarrow 0^-} 3+2x$$

$$= 3+2 \cdot 0$$

$$= 3$$

and, R.H.L = $\lim_{x \rightarrow 0^+} f(x)$

$$= \lim_{x \rightarrow 0^+} 3-2x$$

$$= 3-2 \cdot 0$$

$$= 3$$

As $L.H.L. = R.H.L.$

So the given limit exists at $x=0$ (Ans)

Again for functional value at $x=0$

$$f(x) = 3 - 2x$$

$$= 3 - 2 \cdot 0$$

$$= 3$$

as $L.H.L. = R.H.L. = f(0)$

so the given function is continuous at $x=0$.
(Ans)



Differentiability

1. Definition of Differentiability.

2. $LHD = L f'(c)$

$$= \lim_{h \rightarrow 0^-} \frac{f(c+h) - f(c)}{h}$$

3. $RHD = R f'(c)$

$$= \lim_{h \rightarrow 0^+} \frac{f(c+h) - f(c)}{h}$$

4. If $L.H.D = R.H.D$ then the given function will be differentiable at $x=c$

* Q. Test the differentiability of the function,

$$f(x) = \begin{cases} 5x-4, & 0 \leq x \leq 1 \\ 4x^2-3x, & 1 < x < 2 \end{cases}$$

at the point $x=1$

Solution: Given the function,

$$f(x) = \begin{cases} 5x-4, & 0 \leq x \leq 1 \\ 4x^2-3x, & 1 < x < 2 \end{cases}$$

Now, LHD = $\lim_{h \rightarrow 0^-} \frac{f(c+h) - f(c)}{h}$

$$= \lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h} \quad [\because x=c=1]$$

$$= \lim_{h \rightarrow 0^-} \frac{5(1+h) - 4 - (5 \cdot 1 - 4)}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{5 + 5h - 4 - 1}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{5h}{h}$$

$$\underline{L.H.D = \lim_{h \rightarrow 0^-} 5}$$

$$= 5$$

$$\text{Again, RHD} = Rf'(e)$$

$$= \lim_{h \rightarrow 0^+} \frac{f(e+h) - f(e)}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{4(1+h) - 3(1+h) - (5-1-4)}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{4 + 8h + 4h - 3 - 3h - 1}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{4h + 5h}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{h(4h+5)}{h}$$

$$= \lim_{h \rightarrow 0^+} 4h + 5$$

$$= 4 \times 0 + 5$$

Indeterminate Form (অনির্দিষ্ট আকার)

* Evaluate the limit $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} + 2 \sin x - 4x}{x^5}$
using L'Hospital's rules.

Solution : Given the limit,

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} + 2 \sin x - 4x}{x^5}$$

As $\frac{0}{0}$ form so applying L'Hospital's rules,

we get,

$$\lim_{x \rightarrow 0} \frac{e^x + e^{-x} + 2 \cos x - 4}{5x^4}$$

Again $\frac{0}{0}$ form so applying L'Hospital's rules

we get,

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2 \sin x}{20x^3}$$

Again, as $\frac{0}{0}$ form so applying L'Hospital rules

we get,

$$\lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2\cos x}{60x^2} = \frac{0}{0}$$

Again, as $\frac{0}{0}$ form so applying L'Hospital rules.

we get

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} + 2\sin x}{120x}$$

Again, as $\frac{0}{0}$ form so applying L'Hospital rules.

we get,

$$\lim_{x \rightarrow 0} \frac{e^x + e^{-x} + 2\cos x}{120}$$

$$= \lim_{x \rightarrow 0} \frac{e^0 + e^0 + 2\cos 0}{120}$$

$$= \frac{1+1+2}{120}$$

$$= \frac{1}{30} \text{ (Ans)}$$